

Name: _____ ID: _____ Section: _____

Question 1 [10 Points]

Write a Java method that takes an array as a parameter and moves all the zeros in the array to the start while maintaining the relative order of other elements. Then it returns the given array in a format which matches with the sample output.

You are not allowed to create any additional array to solve the problem. The operations must be performed in-place. The method should work for any given integer array.

Sample Method Call	Sample Output
<code>int [] arr = {2, 0, 6, 1, 0, 4, 0}; System.out.println(moveZeros(arr));</code>	<code>[0, 0, 0, 2, 6, 1, 4]</code>

Solution

```
import java.util.Arrays;
public class MainClass {
    public static void main(String[] args) {
        int[] arr = {6, 0, 1, 3, 0, 7, 0};
        System.out.println(moveZeros(arr));
    }
    public static String moveZeros(int [] arr){
        int zeroIndex = arr.length - 1;

        for (int i = arr.length - 1; i >= 0; i--) {
            if (arr[i] != 0) {

                int temp = arr[zeroIndex];
                arr[zeroIndex] = arr[i];
                arr[i] = temp;

                zeroIndex--;
            }
        }
        return Arrays.toString(arr);
    }
}
```